

Java Assignment 1 – Basic Programs

Part A – Data Types, Variables and Operators

1. Arithmetic Operations

Write a Java program to declare two integer variables and perform **addition, subtraction, multiplication, division, and remainder** operations. Display all the results.

2. Area of a Circle

Write a Java program to store the radius of a circle in a variable and calculate its **area and circumference**.

Formula:

```
Area =  $\pi \times r \times r$   
Circumference =  $2 \times \pi \times r$ 
```

3. Simple Interest

Write a Java program to accept principal amount, rate of interest and time using variables and calculate the **simple interest and total amount**.

Formula:

```
Simple Interest =  $(P \times R \times T) / 100$   
Amount =  $P + \text{Simple Interest}$ 
```

4. Temperature Conversion

Write a Java program to convert temperature from **Celsius to Fahrenheit**.

Formula:

```
Fahrenheit =  $(\text{Celsius} \times 9 / 5) + 32$ 
```

5. Calculate Total and Average

Write a Java program to store marks of **three subjects** in variables and calculate the **total and average marks**.

6. Even or Odd

Write a Java program to accept an integer and check whether the number is **even or odd** using if-else.

7. Positive, Negative or Zero

Write a Java program to accept an integer and check whether it is **positive, negative, or zero** using `if` and `if-else`.

8. Largest of Two Numbers

Write a Java program to accept two numbers and find the **larger number** using `if-else`. If both numbers are equal, display an appropriate message.

9. Voting Eligibility

Write a Java program to accept a person's age and check whether the person is **eligible to vote**.

```
Age >= 18 → Eligible  
Age < 18  → Not Eligible
```

10. Student Result

Write a Java program to accept marks of a student and check whether the student has **passed or failed**.

```
Marks >= 40 → Pass  
Marks < 40  → Fail
```